

AI/ML Internship – Day 31 Notes & Tasks

Module 6: Advanced Matplotlib & Subplots

Part 1: What are Advanced Visualizations?

 Advanced visualizations help display:

- Multiple graphs
- Better analysis
- Professional charts

 In AI/Data Science:

- One graph is often not enough
 - Multiple visualizations are needed
-

Part 2: Theory Answers (IMPORTANT)

1. What is a Subplot?

 A subplot means displaying multiple graphs inside one figure/window.

2. Why Subplots are Used?

 Subplots help:

- Compare multiple datasets
 - Save space
 - Improve analysis
-

3. What is Figure Size?

 Figure size controls graph dimensions.

4. What is Legend?

👉 Legend explains graph labels and plotted data.

◆ 5. Why Advanced Visualization is Important?

👉 Helps create professional dashboards and analysis reports.

📌 Part 3: Creating Multiple Plots

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4]
```

```
y1 = [10, 20, 30, 40]
```

```
y2 = [40, 30, 20, 10]
```

```
plt.plot(x, y1)
```

```
plt.plot(x, y2)
```

```
plt.show()
```

📌 Part 4: Adding Legends

```
plt.plot(x, y1, label="Sales")
```

```
plt.plot(x, y2, label="Profit")
```

```
plt.legend()
```

```
plt.show()
```

📌 Part 5: Figure Size

```
plt.figure(figsize=(8, 5))
```

```
plt.plot(x, y1)
```

```
plt.show()
```

👉 Width = 8

👉 Height = 5

📌 Part 6: Subplots

◆ Two Graphs Together

```
plt.subplot(1, 2, 1)
```

```
plt.plot(x, y1)  
plt.title("Graph 1")
```

```
plt.subplot(1, 2, 2)
```

```
plt.plot(x, y2)  
plt.title("Graph 2")
```

```
plt.show()
```

◆ Understanding subplot()

```
plt.subplot(rows, columns, position)
```

Example:

```
plt.subplot(2, 2, 1)
```

👉 Means:

- 2 rows
 - 2 columns
 - graph position = 1
-

📌 Part 7: Histogram

◆ Used for Data Distribution

```
data = [10, 20, 20, 30, 40, 40, 40]
```

```
plt.hist(data)
```

```
plt.show()
```

👉 Shows frequency distribution.

📌 Part 8: Customizing Charts

```
plt.plot(x, y1, marker="o", linestyle="--")
```

```
plt.title("Custom Graph")
```

```
plt.xlabel("X Axis")
```

```
plt.ylabel("Y Axis")
```

```
plt.grid()
```

```
plt.show()
```

📌 Part 9: Save Graph as Image

```
plt.plot(x, y1)
```

```
plt.savefig("graph.png")
```

👉 Saves graph as image file.

📌 Part 10: Real-World Thinking

👉 In AI/Data Science:

- Accuracy graphs
- Model comparison
- Sales dashboards
- Data distribution analysis

👉 Histograms help:

- Detect patterns
- Detect outliers

Part 11: Visualization Example

```
+-----+  
| Graph 1 | Graph 2 |  
+-----+
```

Subplots help compare graphs together.

Tasks for Day 31

Task 1: Theory (Write Answers)

1. What is subplot?
2. Why subplots are important?
3. What is figure size?
4. What is legend?
5. What is histogram?

Task 2: Multiple Graphs

1. Plot two lines
2. Add labels
3. Add legend

Task 3: Figure Customization

1. Change figure size
 2. Add title
 3. Add grid
-

Task 4: Subplot Practice

1. Create 2 subplots
 2. Create 4 subplots
 3. Compare graphs
-

Task 5: Histogram Practice

1. Create histogram
 2. Analyze distribution
 3. Use random dataset
-

Task 6: Save Graphs

1. Save line graph
2. Save histogram
3. Save subplot figure